

Reamy Womodowe Achirobe

Mechanical Engineer • Thermofluids, CFD & Computational Modelling

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Faculty Intern, Ashesi University • Research Consultant, Klerconsult

RESEARCH PROFILE

Mechanical engineer with research at the intersection of computational fluid dynamics, physics-based modelling, and machine learning for energy systems. Experienced in developing and validating numerical simulation frameworks (ANSYS Fluent), building custom neural network architectures from scratch (Python/TensorFlow), and conducting experimental multiphase flow diagnostics. Strong applied-mathematics foundation from MPhil coursework in CFD, fluid machinery, dynamics, and advanced thermodynamics. Currently teaching CFD and Fluid Mechanics at university level while pursuing research in digital twins and physics-informed modelling. Preparing for marine renewable energy specialisation through an upcoming intensive training at Ocean University of China (July 2026).

EDUCATION

MPhil in Mechanical Engineering (Thermofluids and Energy Systems)

2023 – 2025

Kwame Nkrumah University of Science and Technology (KNUST), Ghana

- Award: Best Graduating Student • Fully funded by KEEP Scholarship
- Thesis: *Using Machine Learning and Infrared Thermography to Identify and Classify Defects in Crystalline Photovoltaics in Ghana*
- Key coursework: Computational Fluid Dynamics, Fluid Machinery, Advanced Engineering Thermodynamics, Dynamics, Computer-Aided Analysis and Modelling, Heat and Mass Transfer

BSc in Mechanical Engineering (Second Class Upper Division)

2016 – 2020

Kwame Nkrumah University of Science and Technology (KNUST), Ghana

- Key coursework: Fluid Mechanics I & II, Behaviour of Real Fluids, Hydraulics & Pneumatics, Vibrations, Strength of Materials I & II, Numerical Analysis, Probability & Statistics, Renewable Energy Conversion

PROFESSIONAL TRAINING

Marine New Energy & Advanced Manufacturing Technologies

Jul 2026

Ocean University of China (OUC), Qingdao, China — 18-day intensive programme

- Covers marine renewable energy technologies, offshore wind blade manufacturing (CRRC New Era), deep-sea engineering and aquaculture equipment (CIMC Raffles), and marine resource development (CNOOC)
- Includes classroom lectures, hands-on laboratory sessions, and factory/industry visits at world-leading offshore energy facilities

PUBLICATIONS AND MANUSCRIPTS

Journal Articles

- Achirobe, R.W., Adaramola, M.S., and Mensah, L.D. *SDCNN: A Shallow Dense Convolutional Neural Network for Automated Fault Detection and Diagnosis in Photovoltaic Systems via Infrared Thermography*. **Engineering Applications of Artificial Intelligence** [Under review, 2025].
- Achirobe, R.W. and Osei, D.A. *Computer Vision Centric Digital Twins for Intelligent Mobility: Simulation of Object Detection and Scene Segmentation*. **International Journal of Innovative Research in Computer Technology (IJIRCT)** [Accepted, 2026].

- Achirobe, R.W. et al. *A physics-based digital twin for photovoltaic hotspot degradation coupling thermal conduction with Arrhenius kinetics*. [In preparation, 2026].

Conference Papers

- Afari, A.A., Fiagbe, Y.A.K., Ramde, E.W., Alhassan, S.T., Quansah, E., and Achirobe, R.W. *Dual Axis Solar Tracker with Low Operating Power Consumption*. **International Conference on Sustainable Systems and Infrastructure Development (ICSSID 2026)**, Paper ID: 188 [Submitted, 2026].

RESEARCH AND TEACHING EXPERIENCE

Faculty Intern — CFD and Fluid Mechanics

Jan 2026 – Present

Ashesi University, Berekuso, Ghana

- Designed and deliver CFD (ME 431) and Fluid Mechanics (ME 432) courses end-to-end using ANSYS Fluent 2025 R2, covering external flow over bodies, converging–diverging nozzle flow, and airfoil aerodynamics
- Implemented $k-\omega$ SST and $k-\varepsilon$ turbulence models in laboratory exercises; wrote simulation guides, problem sets, and exam papers
- Took students from zero simulation experience to completing multi-physics ANSYS Fluent analyses; 100% completion rate

Graduate Researcher — ML and Digital Twins for Energy Systems

2023 – 2025

KNUST, Kumasi, Ghana

- Developed a custom CNN (SDCNN) for automated PV fault detection from infrared thermography; achieved 93–95% accuracy across six fault categories; benchmarked against VGG-16, MobileNetV2, and EfficientNet-B0 under 5-fold CV with paired t -tests and Cohen's d
- Built physics-based digital twin coupling thermal conduction with Arrhenius degradation kinetics for hotspot prediction (manuscript in preparation)
- Collaborated with Prof. M.S. Adaramola (NMBU, Norway) and the Energy Commission of Ghana

Research Assistant — DAAD-Funded Multiphase Flow Project

Feb – Sep 2024

Institute of Particle Technology, University of Erlangen-Nuremberg, Germany

- Characterised fluidized bed hydrodynamics using dual-side pressure transducers, high-speed video, and optical diagnostics
- Applied FFT-based signal processing (Python/NumPy) and image analysis (FIJI) for bubble formation and particle agglomeration studies
- Developed and validated predictor schemes for particle agglomeration; standardised measurement protocol adopted by the research group

National Service — Mechanical Systems Engineering

2020 – 2021

Denys Engineers and Contractors, Ghana

- Led installation, commissioning, and maintenance of centrifugal pumps and pipeline systems across multiple client sites

TECHNICAL SKILLS

- **Programming and ML:** Python (TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, SciPy), MATLAB; CNNs, physics-informed ML, surrogate modelling, data pipelines
- **CFD and Simulation:** ANSYS Fluent, SpaceClaim, Discovery; turbulence modelling ($k-\omega$ SST, $k-\varepsilon$); governing-equation modelling; numerical analysis; physics-based digital twins
- **Experimental Methods:** Pressure transducer instrumentation, infrared thermography (FLIR), high-speed imaging, FFT-based signal processing, optical diagnostics (FIJI)

- **Tools:** $\text{\LaTeX}$, SQL, Excel, Git/GitHub; Apple M1 + TensorFlow-Metal development environment

HONOURS AND AWARDS

- Best Graduating Student — MPhil Mechanical Engineering, KNUST (2025)
- KEEP Scholarship — Full funding for MPhil studies, KNUST (2023–2025)
- DAAD Research Internship Grant — University of Erlangen-Nuremberg, Germany (2024)

LANGUAGES

English (Fluent — sole medium of instruction throughout all university studies) • Kasem (Native) • Twi (Fluent)

PROFESSIONAL AFFILIATIONS

- Vice President, IAESTE KNUST (2024–2025) — Coordinated international engineering trainee placements
- Student Member, Ghana Institution of Engineers (2025)
- Student Member, National Society of Black Engineers (2025)

REFERENCES

Prof. Muyiwa S. Adaramola

Professor, Dept. of Energy and Process Engineering, Norwegian University of Life Sciences (NMBU), Ås, Norway

MPhil co-supervisor and corresponding author • muyiwa.adaramola@nmbu.no

Dr. Lena Dzifa Mensah

Associate Professor, Dept. of Mechanical Engineering, KNUST, Ghana

Principal MPhil supervisor • ldmensah.coe@knust.edu.gh

Dr. Heather Beem

Assistant Professor, Dept. of Engineering, Ashesi University, Ghana

CFD teaching mentor and fluid mechanics pedagogy advisor • hbeem@ashesi.edu.gh